

Document Control

- System Name: Application

Executive Summary

- This Business Requirements Document (BRD) defines all requirements for the Application system, focusing on ingesting customer and order CSV data into Delta tables.
- The document outlines processes for cleansing, transforming, and aggregating data, including SCD Type 2 handling and pipeline orchestration using Delta Live Tables.

Business Objectives

- Load source customer and order CSV files into Delta tables.
- Cleanse data by removing "Null" values and duplicates.
- Compute TotalAmount for orders by multiplying PricePerUnit and Qty.
- Join customer and order data to form an ordersummary table with SCD Type 2 history for customers.
- Aggregate customer spend per day in the customeraggregatespend table.
- Implement the pipeline via Delta Live Tables.

In-Scope

- Ingestion of customer and order CSVs from specified file paths.
- Creation and management of Delta tables: customer, order, ordersummary, customeraggregatespend.
- Data cleansing operations.
- Calculation logic for TotalAmount.
- Join processing and SCD Type 2 implementation for customer data.
- Aggregation of customer spend by Name and Date.
- All orchestration via Delta Live Tables.

Out-of-Scope

- Integration with any data sources, systems, or data fields not listed in the source content.
- Security, user access, SLAs, performance tuning, monitoring, and alerts.
- Scheduling, orchestration, or DevOps details beyond implementation in Delta Live Tables.
- Data quality metrics beyond specified cleansing steps.
- User interface, reporting, and analytics consumption layers.

Stakeholders

- Information not present in source content.

Business Requirements

BR-APP-001 — Source Ingestion

- Description: Load source CSVs into Delta tables for customer and order data from specified paths.
- Business Value: Enables structured ingestion and consistency of data for downstream processes.
- Priority: High

BR-APP-002 — Data Schema Enforcement

- Description: Enforce defined schemas for customer and order tables.
- Business Value: Ensures data integrity and enables trusted analytics.
- Priority: High

BR-APP-003 — Computed Attribute Creation

- Description: Compute TotalAmount in order table as $\text{PricePerUnit} \times \text{Qty}$.
- Business Value: Provides necessary metric for aggregation and reporting.
- Priority: High

BR-APP-004 — Data Cleansing

- Description: Remove records with "Null" values and duplicates in customer and order tables.
- Business Value: Improves data quality and reliability.
- Priority: High

BR-APP-005 — ordersummary Table Creation

- Description: Join customer and order tables on CustId and implement SCD Type 2 for customer changes.
- Business Value: Maintains historical accuracy for analytic reporting.
- Priority: High

BR-APP-006 — SCD Type 2 Maintenance

- Description: Ensure previous customer versions are marked Inactive and new versions Active; manage StartDate and EndDate.
- Business Value: Supports historical data tracking and auditability.
- Priority: High

BR-APP-007 — Aggregate Table Creation

- Description: Create customeraggregatespend table aggregating spend per customer per date.
- Business Value: Delivers actionable spend analytics for business stakeholders.
- Priority: High

BR-APP-008 — End-to-End Pipeline Automation

- Description: Implement the above business requirements using Delta Live Tables.
- Business Value: Enables automated, reliable, and maintainable data processing.
- Priority: High

Assumptions & Constraints

- No additional assumptions are made beyond content provided.
- Only Delta Live Tables to be used for implementation.
- Specific definitions for SCD tracking fields (StartDate, EndDate, Active) are not provided.

Success Metrics

- All target Delta tables are created and populated correctly per requirements.
- Cleansing removes 100% of duplicate and "Null" records.
- ordersummary contains accurate SCD Type 2 customer history.
- customeraggregatespend produces daily spend totals for all customers without data loss.

Risks & Mitigations

- Potential ambiguity due to unspecified data types or SCD attribute fields; mitigate by clarifying with stakeholders before implementation.
- Data quality beyond defined cleansing rules is not addressed; propose proactive data profiling.
- Lack of details for error handling or monitoring; recommend reviewing in Technical design.

Approval

- Approval section to be completed based on stakeholder input.

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
Customer source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded to customer table
Order source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded to order table

Appendix B: Target Definitions

	Layer	Table	Columns	Notes
	Bronze	customer	CustId, Name, EmailId, Region	Loaded
	Bronze	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Loaded

y	Silver	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	SCD T
egatespend	Gold	customeraggregatespend	Name, TotalAmount, Date	Aggreg

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Remove Nulls and duplicates	customer, order	cleansed tables	Drop records with "Null" or duplicate values
2	Calculate TotalAmount	order	order	TotalAmount = PricePerUnit × Qty
3	Join with SCD Type 2	customer, order	ordersummary	Join on CustId, preserve history
4	Aggregate spend	ordersummary	customeraggregatespend	Group by Name, Date; sum TotalAmount

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-1	Aggregation	For each customer per Date, sum TotalAmount	Ordersummary
BR-2	SCD Type 2	Maintain history with StartDate, EndDate, Active fields	Customer changes
BR-3	Cleansing	Remove "Null"/Null values and duplicates	Ingestion

Appendix E: Process Flow

Step	Description	Dependency
1	Load customer and order CSVs to Delta tables	Source files
2	Cleanse customer and order data	Load step
3	Calculate TotalAmount in orders	Cleanse step
4	Join and create ordersummary with SCD Type 2 logic	Calculation step
5	Aggregate daily spend to customeraggregatespend	ordersummary

Appendix F: Entity-Relationship Diagram (ERD)

[customer]

*CustId

Name

EmailId

Region

[order]

*OrderId

ItemName

PricePerUnit

Qty

Date

CustId

TotalAmount

[ordersummary]

*CustId

Name

EmailId

Region

OrderId

ItemName

PricePerUnit

Qty

Date

StartDate

EndDate

Active

[customeraggregatespend]

*Name

TotalAmount

Date

customer 1--* order

customer 1--* ordersummary

order *--1 ordersummary

ordersummary *--* customeraggregatespend